PARISHRAM 2.0 2026 (Class 12th)

Maths

DPP: 3

Matrices

Q1 If $A=\begin{bmatrix}1 & 2\\ -1 & 3\end{bmatrix}$ then $(A^T)^T$ is equal to

(A) $\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}$ (B) $\begin{bmatrix}4 & 0\\ 1 & 5\end{bmatrix}$

(C) $\begin{bmatrix}2 & 0\\ 1 & -2\end{bmatrix}$ (D) $\begin{bmatrix}1 & 2\\ -1 & 3\end{bmatrix}$

Q2 If $A=\begin{bmatrix}1 & 2\\ -1 & 3\end{bmatrix}$ then $(4A)^T$ is equal to

(A) $\begin{bmatrix}4 & 8\\ -4 & 12\end{bmatrix}$ (B) $\begin{bmatrix}4 & -4\\ 8 & 12\end{bmatrix}$

(C) $\begin{bmatrix}5 & 6\\ 3 & 7\end{bmatrix}$ (D) $\begin{bmatrix}16 & 0\\ 4 & 20\end{bmatrix}$

Q3 If $A=\begin{bmatrix}1 & 2\\ -1 & 3\end{bmatrix}$, $B=\begin{bmatrix}4 & 0\\ 1 & 5\end{bmatrix}$. then $(AB)^T$ is equal to

(A) $\begin{bmatrix}6 & 10\\ -1 & 15\end{bmatrix}$ (B) $\begin{bmatrix}6 & -1\\ 10 & 15\end{bmatrix}$

(C) $\begin{bmatrix}6 & 15\\ 1 & 10\end{bmatrix}$ (D) $\begin{bmatrix}15 & -10\\ 1 & 6\end{bmatrix}$

Q4 If $A=\begin{bmatrix}5 & x\\ y & 0\end{bmatrix}$ and $A=A^T$, then

(A) x = 0, y = 5

(B) x + y = 5

(C) x = y

(D) None fo these

Q5 If A is a matrix of order m × n and B is a matrix such that AB^T and B^TA both are defined, then order of matrix B is

(A) m × m (B) n × n

(C) m × n (D) n × m

Q6 For any matrices $\mathbf{A}$ and $\mathbf{B}$ of suitable orders. Which of the following is wrong?

(A) $(\mathbf{A}')'=\mathbf{A}$

(B) $(\mathbf{kA}')'=\mathbf{kA}$(where k is any constant)

(C) $(\mathbf{A}+\mathbf{B})'=\mathbf{A}'+\mathbf{B}'$

(D) $(\mathbf{AB})'=\mathbf{A}'\mathbf{B}'$

Q7 If $\mathbf{A}=\begin{bmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix}$, then $\mathbf{A}+\mathbf{A}'=\mathbf{I}$, if the value of α is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$

(C) π (D) $\frac{3\pi}{2}$

Q8 If $\mathbf{P}$ is a 3 × 3 matrix such that $\mathbf{P}'=2\mathbf{P}+\mathbf{I}$, where $\mathbf{P}'$ is the transpose of $\mathbf{P}$ then

(A) P = I (B) P = – I

(C) P = 2I (D) P = – 2I

Q9 If $\mathbf{A}=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ a & 2 & b\end{bmatrix}$ is a matrix satisfying $AA^T = 9I_3$, then the values of a and b, respectively are:

(A) a = 2, b = 1

(B) a = 2 ,b = – 1

(C) a = – 2, b = 1

(D) a = – 2, b = – 1

Q10 Assertion: If $A=\frac{1}{3}\begin{bmatrix}1 & -2 & 2\\ -2 & 1 & 2\\ -2 & -2 & -1\end{bmatrix}$, then $(A)^TA=I$

Reason: For any square matrix A, $(A^T)^T=A$

(A) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.
(B) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.
(C) Assertion is true; Reason is false.
(D) Assertion is false; Reason is true.

Answer Key

Q1	D
Q2	B
Q3	B
Q4	C
Q5	C
Q6	D
Q7	B
Q8	B
Q9	D
Q10	B

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

$$(A^T)^T = A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

Video Solution:

Q2 Text Solution:

$$(4A)^T = 4A^T = 4\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ 8 & 12 \end{bmatrix}$$

Video Solution:

Q3 Text Solution:

$$AB = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 10 \\ -1 & 15 \end{bmatrix}$$

$$\Rightarrow (AB)^T = \begin{bmatrix} 6 & 10 \\ -1 & 15 \end{bmatrix}^T = \begin{bmatrix} 6 & -1 \\ 10 & 15 \end{bmatrix}$$

Video Solution:

Q4 Text Solution:

As $A = A^T$

$$\Rightarrow \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix} = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}^T$$

$$\Rightarrow \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix} = \begin{bmatrix} 5 & y \\ x & 0 \end{bmatrix}$$

$\Rightarrow x = y$

Video Solution:

Q5 Text Solution:

Let the order of matrix B be p × q. It is given that AB^T and B^TA exist $\Rightarrow \mathbf{n = q}$ and $\mathbf{p = m}$ $\Rightarrow$B is of order $\mathbf{m \times n}$.

Video Solution:

Q6 Text Solution:

As we know a property, $(AB)' = B'A'$.
Therefore, option (D) is wrong.

Video Solution:

Q7 Text Solution:

$$A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$\Rightarrow A' = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$$

Now, $A + A' = I$

$$\therefore \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} + \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$\Rightarrow \begin{bmatrix} 2\cos\alpha & 0 \\ 0 & 2\cos\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Comparing the corresponding elements of the two matrices, we have:

$2\cos\alpha = 1$

$\Rightarrow \cos\alpha = \frac{1}{2} \Rightarrow \alpha = \frac{\pi}{3}$

Video Solution:

Q8 Text Solution:

Given that $P' = 2P + I$

Ta king transpose both the sides, we get

$$\Rightarrow (P')' = (2P + I)'$$
$$\Rightarrow P = 2P' + I$$
$$\Rightarrow P = 2(2P + I) + I$$
$$\Rightarrow 3P = -3I$$
$$\Rightarrow P = -I$$

Video Solution:

Q9 Text Solution:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix} \Rightarrow A^T$$
$$= \begin{bmatrix} 1 & 2 & a \\ 2 & 1 & 2 \\ 4 & -2 & b \end{bmatrix}$$
$$\therefore AA^T = 9I_3$$
$$\Rightarrow \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix} \begin{bmatrix} 1 & 2 & a \\ 2 & 1 & 2 \\ 2 & -2 & b \end{bmatrix}$$
$$= 9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
$$\Rightarrow \begin{bmatrix} 9 & 0 & a+2b+4 \\ 0 & 9 & 2a+2-2b \\ a+2b+4 & 2a+2-2b & a^2+4+b^2 \end{bmatrix}$$
$$= \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$
$$\Rightarrow a+2b+4=0,\ 2a+2-2b=0$$
$$\text{and } a^2+4+b^2=9$$
$$\Rightarrow a+2b+4=0,\ a-b+1=0$$
$$\text{and } a^2+b^2=5$$

Solving $a + 2b + 4 = 0$ and $a - b + 1 = 0$, we get $a = -2$, $b = -1$. Clearly, these values satisfy $a^2 + b^2 = 5$.

Hence, $a = -2$ and $b = -1$.

Video Solution:

Q10 Text Solution:

$$AA^T = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$$
$$\cdot \frac{1}{3}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & 2 & -1 \end{bmatrix}$$
$$= \frac{1}{9}\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$\therefore$ Assertion is true.

Reason is true, as for any matrix A, $(A^T)^T = A$.

∴ Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

Video Solution:

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